# A focus at the sample

Four complete CRL energy sweeps · horizontal knife-edge measurements · SMI

## Two clear foci. A geometry near 1.6 m.

Holders 3 and 4 produce clear, energy-bracketed narrow cores. With the deliberately focused small SSA as the secondary source, their measured best energies support the documented 1600 mm geometry.

**HOLDER 4 · Z = 0**

**4.70 µm**

at 8.77 keV | sampled core FWHM

**HOLDER 3 · Z = +200 mm**

**5.39 µm**

at 16.20 keV | sampled core FWHM

| MEASUREMENT | RESULT | INTERPRETATION |
|---|---|---|
| Holder 3 · 1170759 | 16.20 keV / 5.39 µm | Narrow core; visible wings |
| Holder 4 · 1170765 | 8.77 keV / 4.70 µm | Cleanest new focus |
| Holder 5 · 1170763 | 6.325 keV / 6.38 µm | Core on a broad pedestal |
| Holder 2 · 1170769 | No accepted optimum | Low counts; moving / clipped edge |

### The setup matters

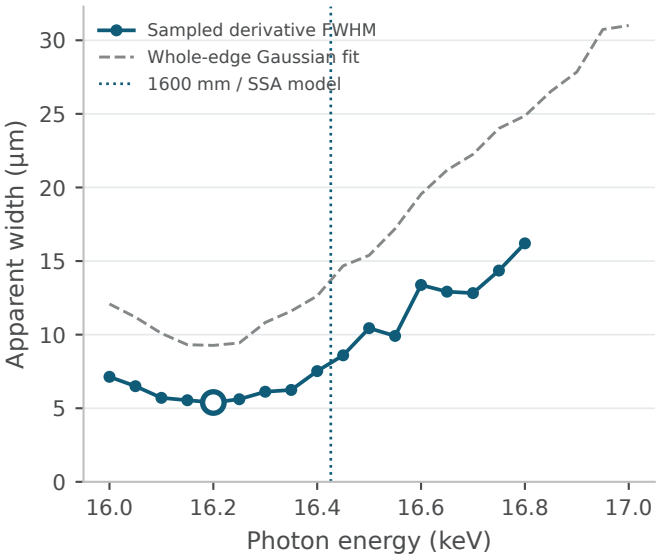
Mirror bending voltages were optimized for maximum flux through the 30 × 49.9 µm SSA aperture. This supports an upstream waist at or near the SSA. Nominal lens powers then imply Z=0 distances of 1557 and 1631 mm; their difference remains unresolved.

Widths are sampled horizontal line-profile measurements; decimals do not imply submicron precision.

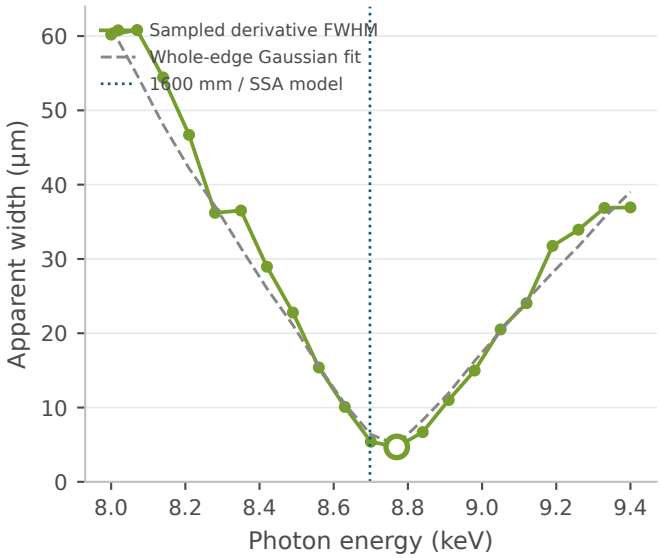
# The four energy sweeps

A common analysis convention, with different levels of physical interpretability.

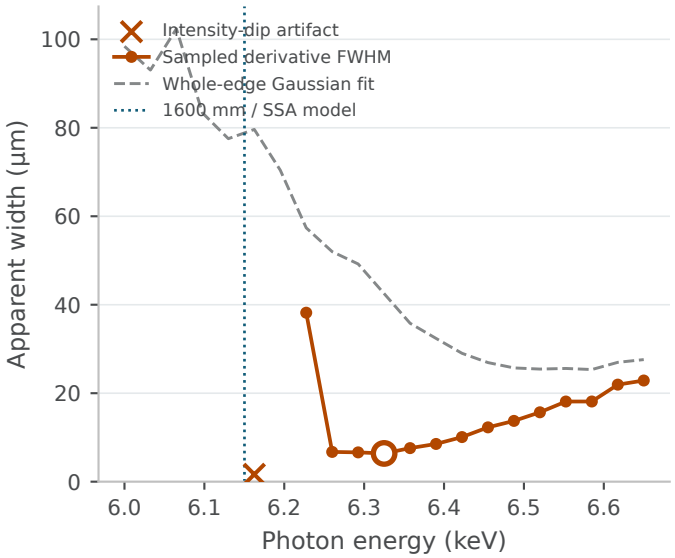
## Holder 3 / 1170759



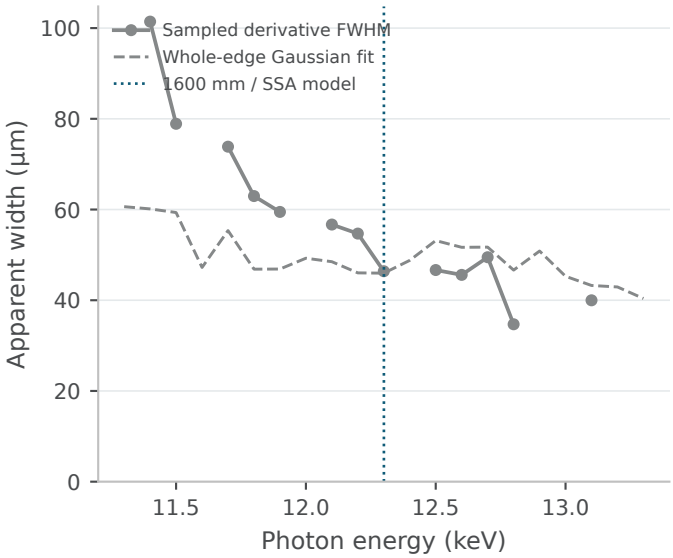

## Holder 4 / 1170765




## Holder 5 / 1170763




## Holder 2 / 1170769



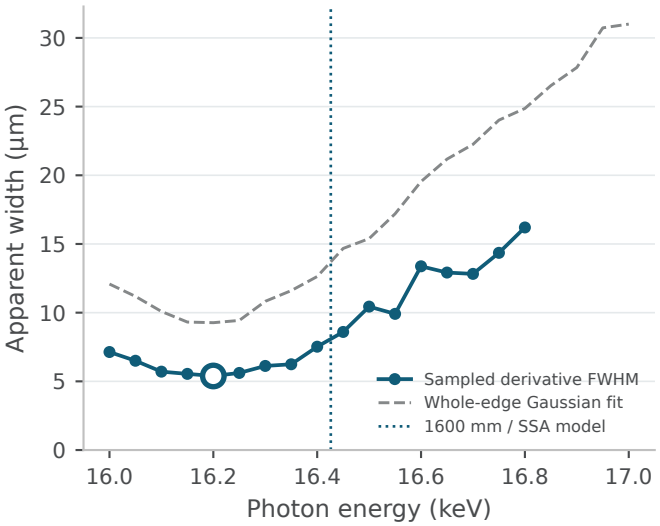

### Read the curves together with the raw profiles

Dashed widths are whole-edge Gaussian diagnostics, not accepted optical spot sizes. Holder 5's crossed-out minimum is an intensity artifact. All holder-2 widths are diagnostic; no best-focus energy is assigned.

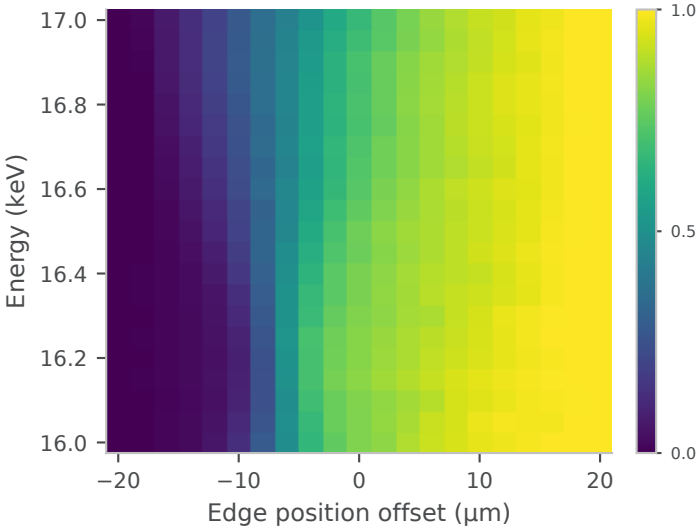
# A narrow core with wings

Holder 3 · 16 nominal R50 µm Be elements · CRL z = +200 mm

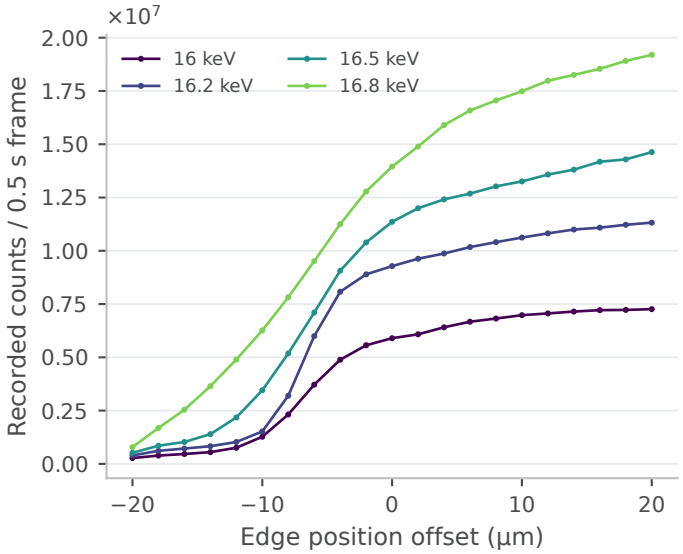
## 01 Focus versus energy



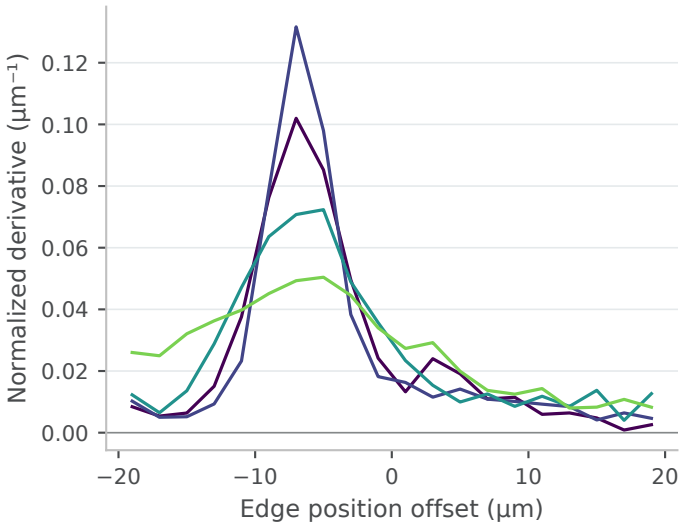

## 02 Every measured edge




## 03 Selected raw profiles




## 04 Profile shape / derivative




Position origin: -14698.00 µm (fixed for this run). Map and derivatives use endpoint normalization;
0 and 1 denote the first/last three-point means, not proven closed/open plateaus. Map color clips at 0–1.

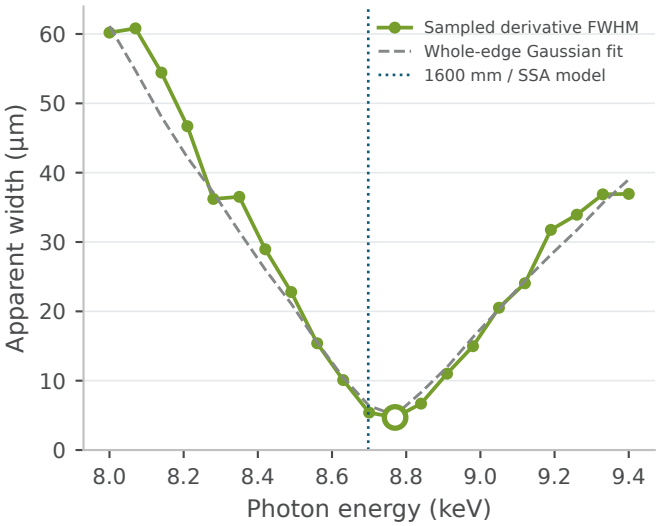
### 5.39 µm at 16.20 keV

Neighboring widths are 5.54 and 5.61 µm at 16.15 and 16.25 keV. The whole-edge Gaussian width is 9.27 µm at the minimum, exposing broader wings. Above 16.80 keV, half-height coverage fails. Harmonic 13 changes to 15 at 16.15 keV.

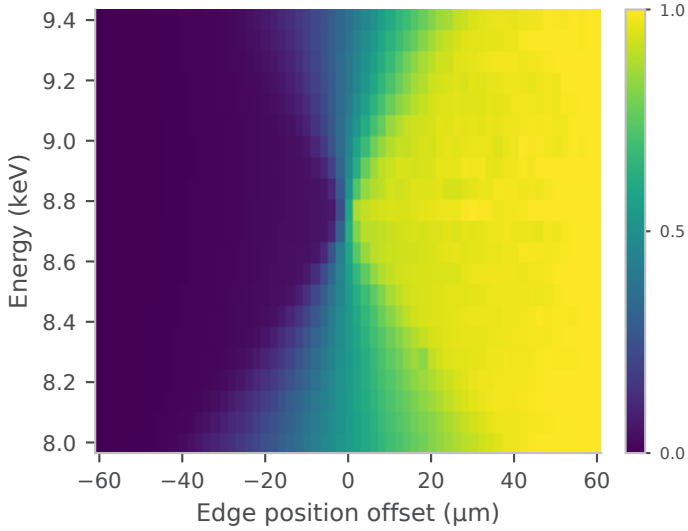
# The cleanest new focus

Holder 4 · 4 nominal R50 µm Be elements · CRL z = 0 mm

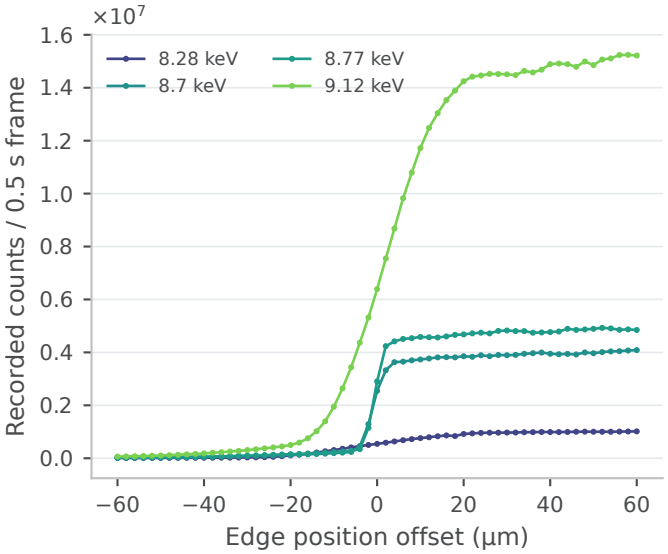
## 01 Focus versus energy



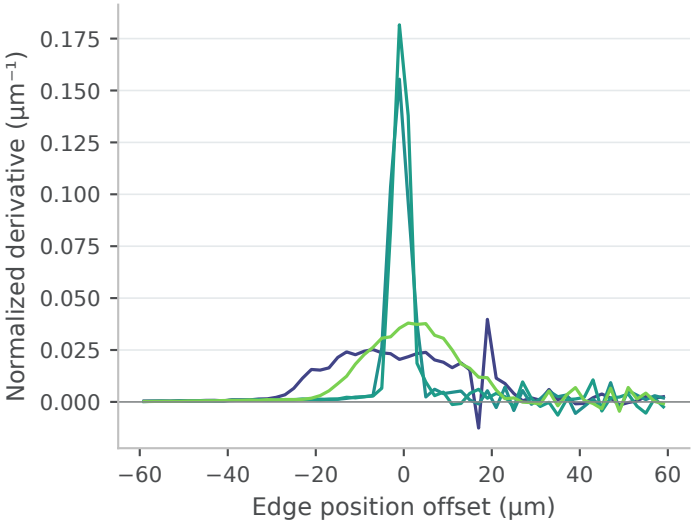

## 02 Every measured edge




## 03 Selected raw profiles




## 04 Profile shape / derivative




Position origin: -14670.04 µm (fixed for this run). Map and derivatives use endpoint normalization;
0 and 1 denote the first/last three-point means, not proven closed/open plateaus. Map color clips at 0–1.

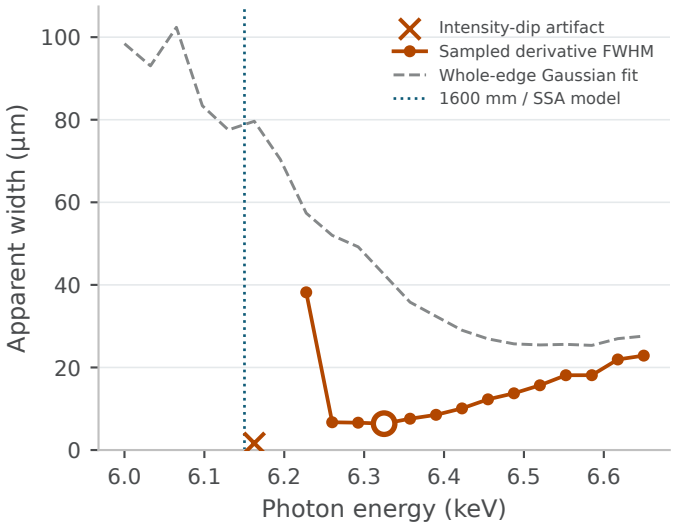
### 4.70 µm at 8.77 keV

Widths broaden on both sides of the minimum: 5.40 µm at 8.70 and 6.69 µm at 8.84 keV. The whole-edge Gaussian width is 5.03 µm at 8.77 keV. Harmonic 7 remains fixed. This is the strongest new evidence for near-nominal lens power.

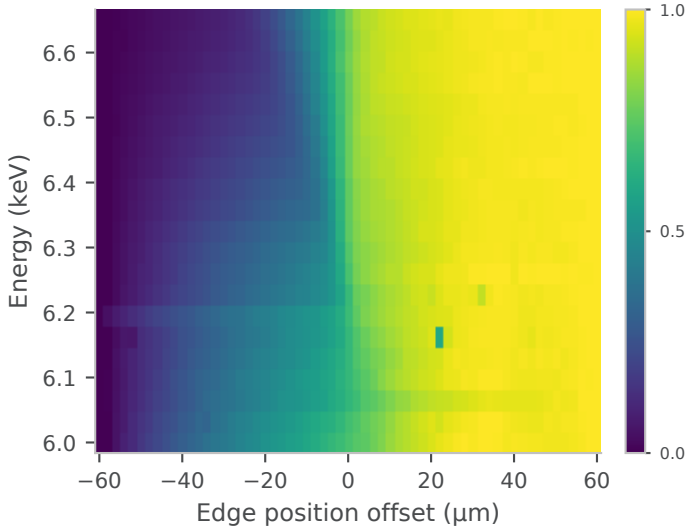
# A focused core, a broad pedestal

Holder 5 · 2 nominal R50 µm Be elements · CRL z = 0 mm

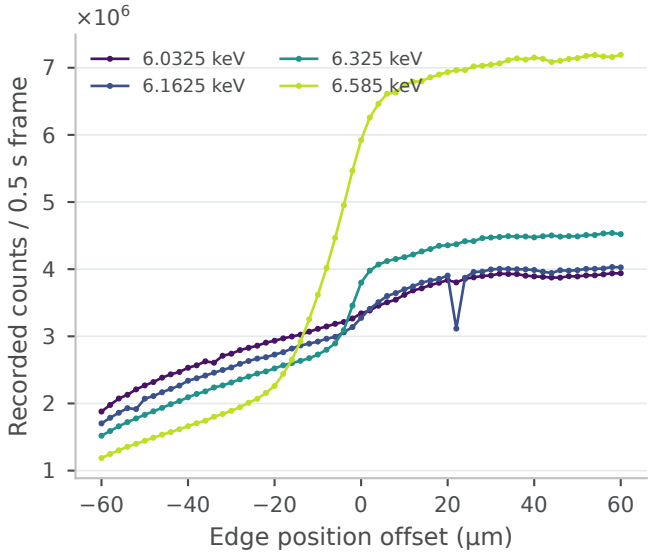
## 01 Focus versus energy



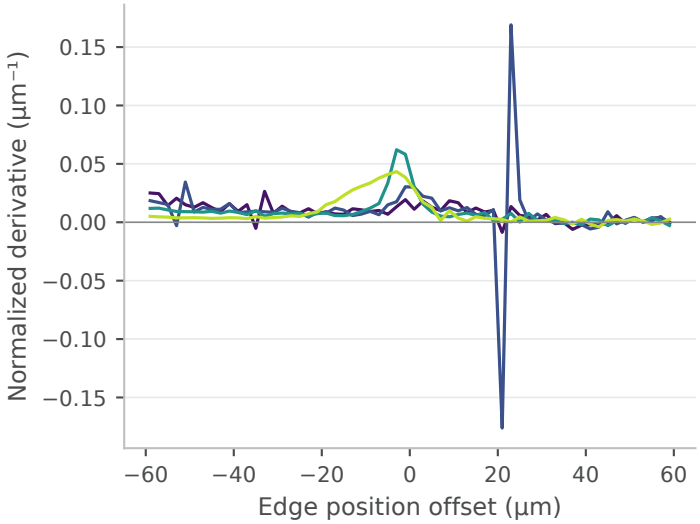

## 02 Every measured edge




## 03 Selected raw profiles




## 04 Profile shape / derivative




Position origin: -14679.05 µm (fixed for this run). Map and derivatives use endpoint normalization;
0 and 1 denote the first/last three-point means, not proven closed/open plateaus. Map color clips at 0–1.

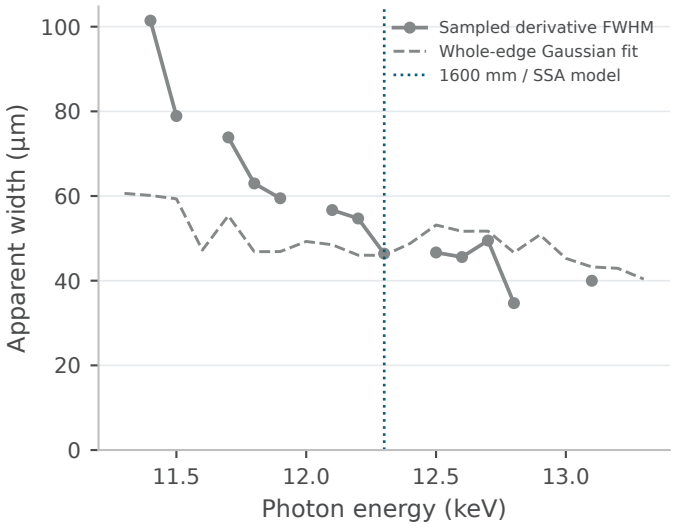
### Core minimum near 6.325 keV

The core reaches 6.38 µm, but the whole-edge width there is 42.5 µm. Broad-profile widths instead bottom near 6.5–6.6 keV. At 6.1625 keV, a large intensity drop and recovery creates a false 1.65 µm derivative minimum; it is not a focus.

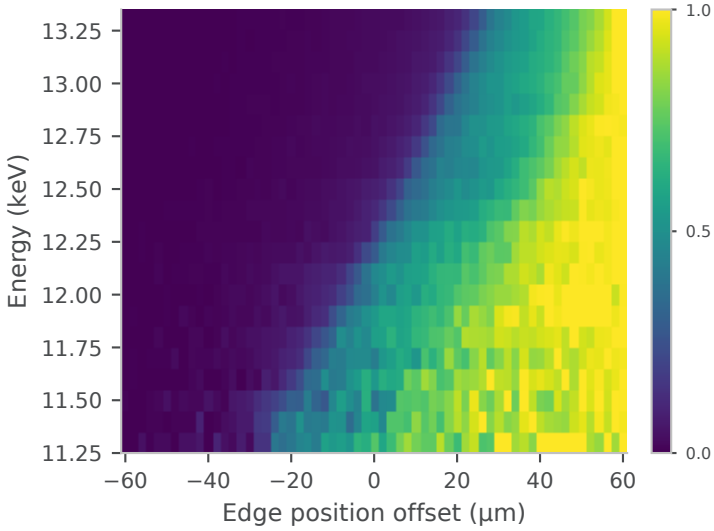
# An unresolved alignment

Holder 2 · 8 nominal R50 µm Be elements · CRL z = 0 mm

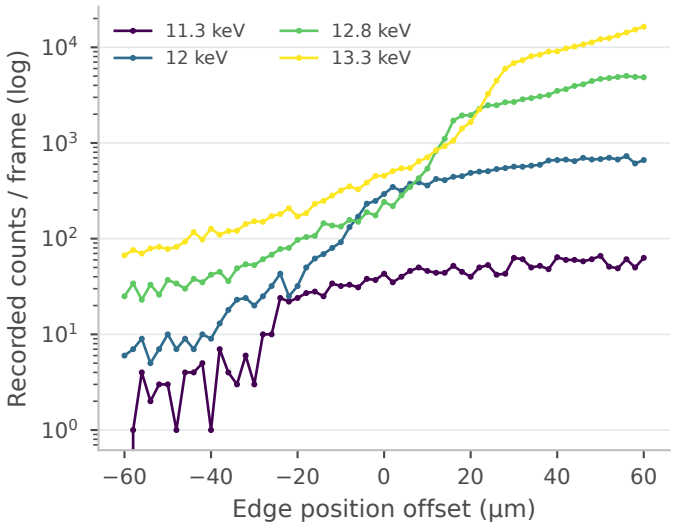
## 01 Focus versus energy



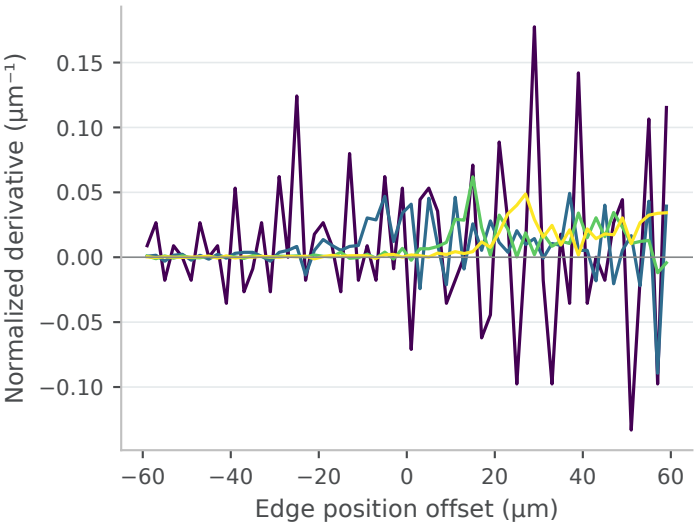

## 02 Every measured edge




## 03 Selected raw profiles




## 04 Profile shape / derivative




Position origin: -14667.96 µm (fixed for this run). Map and derivatives use endpoint normalization;
0 and 1 denote the first/last three-point means, not proven closed/open plateaus. Map color clips at 0–1.

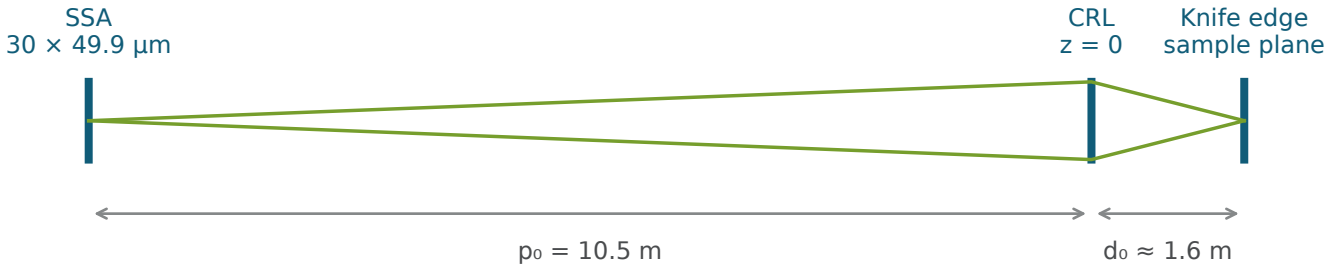
### No defensible best-focus energy

The fitted edge center shifts by about 50 µm across the sweep. Open-side counts rise from roughly 58 to 15,300 per frame, while the high-energy tail is clipped. The automatic 34.7 µm minimum at 12.8 keV is not a reliable focus calibration.

# What the focus says about distance

Primary model: nominal two-sided Be lens power, with the effective secondary source at the SSA.

## Mirror bending optimized for transmission through the small SSA




| HOLDER / Z | MEASURED | 1600 mm + SSA | 1350 mm + SSA |
|---|---|---|---|
| 3 / +200 mm | 16.20 keV | 16.426 keV | 15.044 keV |
| 4 / 0 mm | 8.77 keV | 8.698 keV | 8.073 keV |

Inferred Z=0 distance: 1557 mm (holder 3) versus 1631 mm (holder 4).

## Near-nominal power; not yet a precision distance calibration

At fixed 1600 mm geometry, the clean measurements imply effective lens-power scales of 0.973× nominal (holder 3) and 1.017× (holder 4). The agreement is at the few-percent level, but the holder-to-holder distance discrepancy is larger than the neighboring-energy sampling brackets. Principal-plane offsets, actual radii, alignment and the width metric remain relevant.

### The spot-size scale also makes sense

Geometrically imaging a fully illuminated 30 µm horizontal SSA gives about 4.57 µm at z=0 and 3.93 µm at z=+200 mm. The measured cores lie on this few-micron scale. These aperture-image extents are not Gaussian FWHM predictions.

Sensitivity only: collimated input would imply ~1.40–1.41 m. The known SSA-focusing procedure favors the SSA model; focus energies alone cannot independently determine distance, curvature and lens strength.

# A measurement record you can revisit

Tiled smi/raw · scalar counts and motor positions · acquisition configuration and baseline snapshots

## Acquisition

All four complete scans: 21 energies, 0.5 s exposure and 2 µm edge steps. The initial scan spans 40 µm (441 frames); new scans span 120 µm (1281 frames each). SSA gaps and mirror bending settings are consistent across acquisition snapshots. CRL angles differ between holders.

## How the widths were measured

Sampled core width: unsmoothed positive first differences, then model-free derivative FWHM using smi_plans.analysis.analyze_xy. Both endpoints must be below half height. Raw-profile inspection is essential: this test alone does not reject intensity spikes, noise or multi-component beams. Dashed curves show an independent cumulative-Gaussian edge fit; they are diagnostic, not a substitute for core width.

### Interpretation limits

A 4.7 µm width spans only ~2.4 steps. These are horizontal knife-edge widths, without deconvolution or a vertical measurement. Count rates are not calibrated flux or lens transmission. Per-frame detector sensitivity readbacks are absent from the saved primary fields.

| PARTIAL RUN | RECORDED / PLANNED | USE IN THIS REPORT |
|---|---|---|
| 1170761 · holder 5 | 105 / 1281 | Attenuation / signal-loss diagnostic |
| 1170767 · holder 2 | 1037 / 1281 | Failed pitch recovery; corroborating data |
| 1170771 · holder 3 | 53 / 1281 | Incomplete first edge; no focus minimum |

## Reproducibility & next step

Companions: MEASUREMENT_REPORT.md, measurement_report.html, measurement_edges.csv, measurement_geometry.csv and measurement_runs.csv. The measurement_cache folder preserves scalar data and full run UIDs for 13 runs, including six coarse scans. This PDF is generated offline by make_measurement_pdf.py; it does not access hardware. Next: repeat clean holder 4 at several Z positions under the same small-SSA illumination to refine the geometry.